

Olist Brazilian E-Commerce: Delivery Performance and Customer Satisfaction Analysis

A Data Analytics Portfolio Project

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Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle) — 100k orders, 2016–2018

Report status: Business Problem, Assumptions, Research Questions, and Hypotheses are finalized below.
Analysis, Findings, and Suggestions will be added after the data analysis is completed.

Business Problem

Olist connects small and medium sellers across Brazil to multiple online marketplaces and handles logistics through partner carriers. Customer reviews and repeat purchases directly affect seller and platform revenue, and the biggest complaint in e-commerce logistics is usually delivery time. Some deliveries arrive well past the estimated date, and it is not clear how much this actually hurts customer satisfaction, or which sellers, product categories, states, and time periods are most affected.

Understanding this relationship would let Olist target the specific weak points in its logistics network instead of making broad, unfocused changes.

The analysis of order delivery timing, freight cost, product category, and seller/customer location, and how each relates to review scores and repeat purchase behavior, is the main focus of this report.

Assumptions

1. Only orders with status “delivered” are used for delivery-time analysis, since cancelled or unavailable orders have no real delivery date to measure.
2. The review score reflects the overall purchase experience, and delivery timing is one of its main drivers, though not the only one.
3. Data from 2016 is too small a sample (the platform was just starting) to draw reliable monthly comparisons, so 2017 to mid-2018 is treated as the main analysis window.
4. `customer_unique_id` represents one real person across multiple orders; `customer_id` changes per order and is not used for repeat-purchase analysis.
5. The estimated delivery date given to the customer at checkout is treated as the promise Olist needs to keep, regardless of why an order was delayed.
6. Sellers are assumed to operate under similar platform rules, so a comparison across sellers is fair for this analysis.
7. No unusual one-time events (strikes, holidays, courier issues) in 2017 to 2018 are assumed to have caused permanent shifts in delivery patterns, only temporary spikes.

Research Questions

1. What factors are associated with late deliveries?
2. How much do late deliveries reduce customer review scores?
3. Which states, sellers, or product categories have the worst delivery performance?
4. Does a late first delivery reduce the chance a customer orders again?
5. What should Olist change to reduce delays and protect satisfaction?

Hypotheses

H1. Orders delivered after the estimated delivery date receive lower review scores than orders delivered on time.

H2. Orders where the seller and customer are in different states take longer to deliver than same-state orders.

H3. A higher freight cost relative to product price is associated with lower review scores.

H4. Late delivery rates are not the same across all states, and are higher in certain months (e.g. around November, due to Black Friday volume).

H5. Customers whose first order arrives late are less likely to place a second order than customers whose first order arrives on time.

Analysis and Findings, and Suggestions sections will be appended once the data has been loaded, cleaned, and tested against these hypotheses.